Fingerprint Matching Pipeline Diagnostic Report

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Contents

1 Scope

This report summarizes the diagnostic experiments performed on the contactless fingerprint matching pipeline. The primary objective was to determine why two real captures of the same finger produce a relatively low genuine score, even though synthetic shift and rotation tests show reasonable robustness.

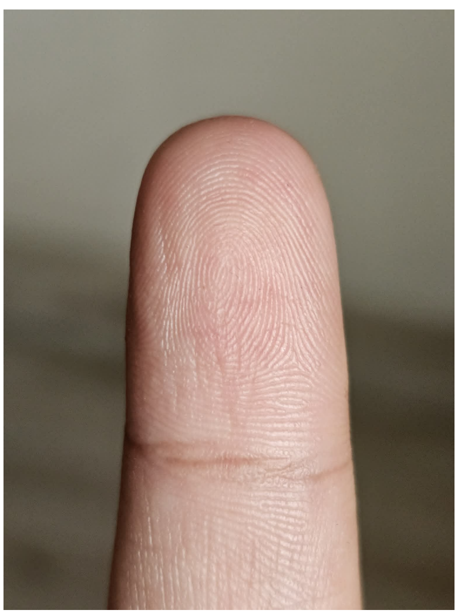
Unless otherwise stated, experiments used the configuration in Table 1.

Table 1: Default diagnostic settings.

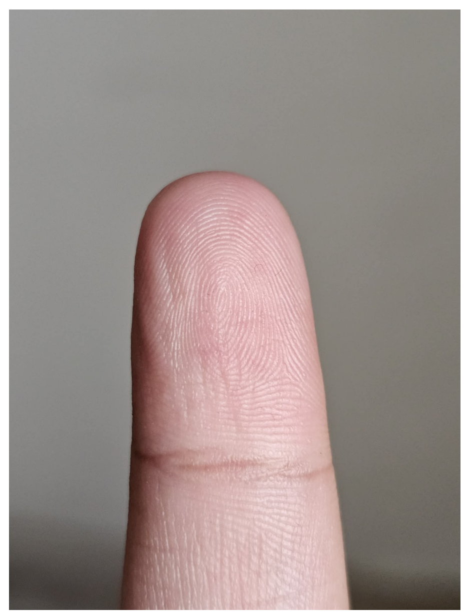
Setting	Value
Model weights	`weights/best.pt`
Matcher	`LSA-CENTROID`
Minutia threshold	`0.6`
NMS	on
Segmentation mode	`finger-pad-auto`
Device	CPU

2 Primary Captures

The main real same-finger pair used throughout the diagnosis is shown in Figure 1.



Capture A



Capture B

Figure 1: Primary contactless captures used for diagnostics. These two pictures are genuine pairs

3 Experiment 1: Segmentation and Masking Failure

3.1 Purpose

The first failure observed was that the old trusted path produced broken masks. The foreground and masked images removed or blacked out large parts of the finger pad, especially in one capture. This made downstream minutia extraction unreliable or impossible.

3.2 Change Made

A new `finger-pad-auto` path was implemented for controlled contactless finger photos.

Table 2: Segmentation-path changes and their effects.

Change	Result
Removed `rembg` from trusted inference path	Avoids generic foreground-removal failures.
Added deterministic finger pad segmentation	Uses skin, geometry, and ridge-energy cues.
Added full-finger and distal-pad quality gates	Causes bad masks to fail before FeatureNet and MCC.
Added repair pass for background-connected masks	Handles warm background strips connected to the finger.
Added debug overlays and rejection artifacts	Makes segmentation failures inspectable.

3.3 Indication

The original segmentation path was a major pipeline risk. The replacement made the pipeline fail-loud and produced usable masks for the controlled captures. However, later experiments show that even with improved masks, real same-finger captures still score low, so segmentation is not the only remaining failure.

4 Experiment 2: Same-Image Determinism and Fixed-Period Diagnostics

4.1 Purpose

This experiment checked whether FeatureNet and decoding are stable when the same image is reused, and then under tiny controlled transforms. Ridge-period scaling was fixed at `13.0846` to remove scale-estimation noise.

4.2 Data

Table 3: FeatureNet decoding counts under same-image and small-transform conditions.

Case	Input shape	Score-map shape	Raw active cells ≥ 0.6	Local maxima	Decoded NMS on	Decoded NMS off
base	1565x1174	195x146	102	50	50	102
same copy	1565x1174	195x146	102	50	50	102
shift x5	1565x1174	195x146	87	46	46	87
rotate +5 deg	1565x1174	195x146	119	53	53	119

Table 4: Counterpart recovery and score-map correlation.

Comparison	NMS on within 8 px	NMS on within 16 px	NMS on within 24 px	NMS on within 32 px	NMS off within 16 px	Score-map correlation
same copy	50/50	50/50	50/50	50/50	50/50	1.0000
shift x5	34/50	36/50	39/50	43/50	36/50	0.9521
rotate +5 deg	12/50	34/50	42/50	47/50	35/50	0.6118

Table 5: Failure categories for small-transform diagnostics.

Comparison	Matched	Model score changed	Threshold drop	NMS suppressed	Offset moved	Orientation changed
same copy	50	0	0	0	0	0
shift x5	36	5	2	0	7	0
rotate +5 deg	31	1	2	1	12	3

4.3 Indication

The same-image path is deterministic. Tiny transforms do change FeatureNet responses and localization, but NMS is not the primary cause. Disabling NMS does not recover most missing counterparts. The dominant instability is in score-map extraction and in the offset and orientation heads.

5 Experiment 3: Same-Image MCC Robustness With Fixed Ridge Period

5.1 Purpose

This experiment measured whether the final MCC matcher remains robust when a single capture is synthetically shifted or rotated. This isolates matcher robustness from real capture variability.

5.2 Data

Table 6: MCC scores for synthetic transforms of the same capture.MCC scores indicates what percentage of the fingerprint matches.

Transform	Minutiae	MCC score
same image	`50 -> 50`	1.0000
shift x10	`50 -> 52`	0.7744
shift x15	`50 -> 64`	0.7408
shift x20	`50 -> 44`	0.9358
rotate +10 deg	`50 -> 54`	0.7716
rotate +15 deg	`50 -> 55`	0.9456
rotate +20 deg	`50 -> 54`	0.8881

5.3 Indication

The matcher is reasonably robust to synthetic transforms of the same capture. Scores are not monotonic: larger transforms sometimes score higher. This indicates branchy behavior in segmentation, preprocessing, and FeatureNet extraction, but the final matcher does not collapse under these synthetic transforms.

The `shift x15` case produced `64` minutiae from a source that normally produced `50`. Inspection showed that this was caused by a changed segmentation/preprocessing branch and a hotter FeatureNet score map, not by MCC or NMS.

6 Experiment 4: Same-Image Combined Shift-Then-Rotate Sweep

6.1 Purpose

This experiment stress-tested robustness when both translation and rotation are applied to the same capture before matching back to the original.

6.2 Settings

Table 7: Combined same-image transform sweep settings.

Setting	Value
Fixed ridge period	`13.0846`
Shifts	`x10` through `x50`
Rotations	`5, 10, 15, 20 deg`
Rotation canvas	same canvas

6.3 Data

Table 8: MCC scores for combined synthetic transforms of the same capture.

Shift	Rot 5	Rot 10	Rot 15	Rot 20
x10	0.7894	0.7808	0.8786	0.9107
x15	0.7607	0.7576	0.7949	0.7144
x20	0.7690	0.7853	0.8484	0.8766
x25	0.7350	0.7444	0.7571	0.7352
x30	0.7965	0.7890	0.8388	0.8567
x35	0.7690	0.7911	0.7611	0.7417
x40	0.6958	0.7738	0.8246	0.8720
x45	0.7324	0.7398	0.7505	0.7724
x50	0.8075	0.7830	0.8091	0.8069

6.4 Indication

The same-capture matcher remains robust under combined synthetic perturbations. This supports the conclusion that the low real-pair score is not simply a transform-invariance failure in MCC.

7 Experiment 5: Real Pair With Transformed Probe Capture

7.1 Purpose

This experiment checked whether the real same-finger pair maintains similar robustness when the second real capture is shifted and rotated before matching against the first real capture.

7.2 Settings

The settings are the same as Experiment 4, except the transformed image is Capture B and the reference remains Capture A.

7.3 Data

Table 9: Baseline real-pair score under fixed-period condition.

Pair	Score	Minutiae
Capture A vs Capture B	0.6167	`50 -> 64`

Table 10: MCC scores for transformed Capture B matched against Capture A.

Shift	Rot 5	Rot 10	Rot 15	Rot 20
x10	0.6333	0.5960	0.6300	0.5750
x15	0.6488	0.6351	0.6534	0.6425
x20	0.6260	0.5890	0.6108	0.6543
x25	0.6331	0.6284	0.6530	0.6155
x30	0.6230	0.5922	0.6368	0.6628
x35	0.6394	0.6390	0.6535	0.6727
x40	0.6034	0.5953	0.6124	0.6557
x45	0.6501	0.6465	0.6611	0.6025
x50	0.6219	0.6473	0.5979	0.6071

7.4 Indication

The real-pair scores remain clustered around `0.58--0.67` under synthetic transforms. That means the transform itself is not the main failure. The real pair already has mediocre correspondence before perturbation.

8 Experiment 6: Forcing Capture B to Use Capture A Preprocessing Scalars

8.1 Purpose

This experiment tested whether the real-pair score is low because Capture B estimates different preprocessing scalar values: ridge period, scale, and yaw.

8.2 Data

Table 11: Real-pair score with and without forced preprocessing scalars.

Run	Score	Minutiae
Normal Capture A vs normal Capture B	0.6624	`50 -> 54`
Normal Capture A vs Capture B forced to use Capture A scale/yaw	0.5708	`50 -> 56`

Table 12: Preprocessing values for normal and forced runs.

Value	Capture A normal	Capture B normal	Capture B forced
ridge period	13.0846	12.0541	13.0846
scale	0.7643	0.8296	0.7643
yaw	-0.4448 deg	-2.5258 deg	-0.4448 deg

8.3 Indication

Forcing Capture B to use Capture A preprocessing scalars made the score worse. Therefore, scale and yaw mismatch is not the primary reason this pair scores low.

9 Experiment 7: Real Same-Finger Failure Waterfall

9.1 Purpose

This experiment diagnosed exactly where correspondence collapses for the primary real pair.

9.2 Data

Table 13: Preprocessing and extraction summary for the primary real pair.

Image	Ridge period	Scale	Yaw	Shape	Mask area	BBox	Minutiae	MCC descriptors	Raw active cells	Local maxima
Capture A	13.0846	0.7643	-0.4448	`1565x1174`	411181	`304,270,600,818`	50	42	102	50
Capture B	12.0541	0.8296	-2.5258	`1699x1274`	388290	`322,451,578,828`	54	51	118	54

Table 14: Alignment and MCC metrics for the primary real pair.

Metric	Value
Final MCC score	0.6624
Similarity matrix shape	`42 x 51`
LSA selected pairs	12
Counterparts within 16 px after diagnostic alignment	`5 / 50`
Counterparts within 32 px after diagnostic alignment	`16 / 50`
Median location error	45.98 px
Median angle error	21.00 deg
Score-map responses found	`17 / 50`
Pre-NMS candidates found	`17 / 50`
Survived NMS	`16 / 50`
Descriptor pairs available	1057

Table 15: Per-minutia failure categories for the primary real pair.

Failure category	Count
model_score_missing	31
offset_moved	11
descriptor_score_low	4
mcc_descriptor_dropped	1
nms_suppressed	1
matched	2

9.3 Indication

The failure occurs mostly before MCC assignment. After diagnostic alignment, only 17/50 Capture A minutiae have a nearby score response in Capture B, and only 16/50 survive NMS. The dominant failure category is model_score_missing, with 31 affected minutiae.

This indicates that FeatureNet extraction and image-quality robustness are the likely failure points for the real same-finger pair. MCC is operating on a weak and inconsistent set of extracted minutiae.

10 Experiment 8: Genuine vs. Impostor Cross-Check

10.1 Purpose

This experiment compared two right-index captures and two right-middle captures in all permutations to evaluate genuine/impostor separation.

10.2 Data

Table 16: Genuine and impostor cross-check scores.

A	B	Expected	Score
ind1	ind1	self	1.0000
ind1	ind2	genuine index	0.6624
ind1	mid1	impostor	0.5484
ind1	mid2	impostor	0.5351
ind2	ind1	genuine index	0.6624
ind2	ind2	self	1.0000
ind2	mid1	impostor	0.5089
ind2	mid2	impostor	0.5055
mid1	ind1	impostor	0.5484
mid1	ind2	impostor	0.5089
mid1	mid1	self	1.0000
mid1	mid2	genuine middle	0.5257
mid2	ind1	impostor	0.5351
mid2	ind2	impostor	0.5055
mid2	mid1	genuine middle	0.5257
mid2	mid2	self	1.0000

Table 17: Score ranges by comparison bucket.

Bucket	Score range
Self controls	1.0000
Genuine index	0.6624
Genuine middle	0.5257
Impostors	0.5055–0.5484

10.3 Indication

Genuine/impostor separation is weak. The middle-finger genuine pair scores inside the impostor range. The index-finger genuine pair separates from impostors, but by a limited margin. This confirms that the issue is not only one pair; real same-finger capture consistency is weak.

11 Overall Findings

Table 18: Summary answers to the diagnostic questions.

Question	Answer
Does the pipeline return perfect self-match?	Yes. Self controls consistently score `1.0`.
Is MCC robust to synthetic shifts/rotations of the same capture?	Mostly yes. Same-capture transformed scores generally stay high.
Is preprocessing scale/yaw mismatch the main issue for Capture A vs Capture B?	No. Forcing Capture B to use Capture A scale/yaw reduced score.
Is NMS the main issue?	No. Disabling NMS did not recover most missing counterparts.
Where does the real-pair correspondence collapse?	FeatureNet score-map extraction and localization. Most Capture A minutiae do not reappear as nearby score responses in Capture B.
Is genuine/impostor separation currently strong?	No. The genuine middle-finger score overlaps impostor scores.

12 Final Conclusion

The current failure point is not primarily MCC assignment, NMS, or scalar preprocessing mismatch. The dominant failure is unstable FeatureNet minutia extraction across real captures of the same finger.

The real-pair waterfall shows that:

- `31/50` Capture A minutiae are classified as `model_score_missing` in Capture B.
- Only `5/50` Capture A minutiae have decoded Capture B counterparts within `16 px` after diagnostic alignment.
- Only `16/50` have counterparts within `32 px`.
- MCC descriptors exist in reasonable counts, `42` and `51`, but they are built from inconsistent minutia sets.